



FEATURES OF THE MOON

From afar, the Moon is clearly divided into two types of terrain. There are large, dark plains called maria (Latin for “seas”) and also brighter, undulating, heavily cratered highland regions. The whole surface was initially covered with craters, most of which were produced during a time of massive bombardment. The rate at which asteroids have been striking the Moon has decreased over the last 4 billion years. Around 4 billion years ago, the Moon was also volcanically active. Lava rose to the surface through cracks and fissures, filling the lower parts of the large craters to produce the dark plains. The plains reflect only about 4 percent of the sunlight that hits them, whereas the mountains reflect about 11 percent.

MOSAIC OF THE NORTH POLE

The lunar north pole is partially hidden from view from Earth and is best imaged by orbiting spacecraft. Galileo took a series of photographs of the region on December 7, 1992, on its way to Jupiter.

NEAR SIDE *northern hemisphere*

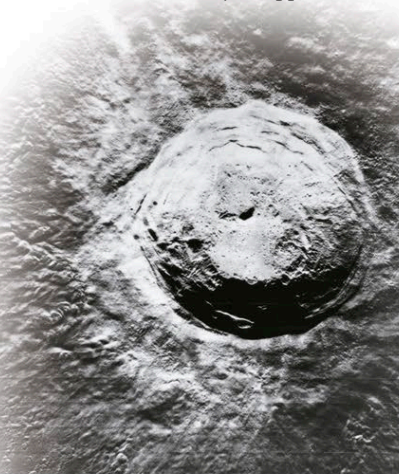
Aristarchus Crater



TYPE Impact crater
AGE About 300 million years
DIAMETER 23 miles (37 km)

This young crater has a series of nested terraces, which were produced by concentric slices of rock in the wall slipping downward. This both widened the crater and made it considerably shallower, as the initially deep central region was filled with material from the rim. Aristarchus was mapped by the Apollo Infrared Scanning Radiometer. During the night, the temperature in the crater

is about 54°F (30°C) higher than that of the surrounding terrain. Young craters contain many large boulders. These take a long time to heat up during the day and also a long time to cool down at night. As time passes, the boulders are broken up by small impacting asteroids, so this thermal difference eventually disappears.



LUNAR ORBITER 5 IMAGE

This view of Aristarchus, taken from directly above, underlines the crater's circularity and reveals the extensive surrounding blanket of hummocky ejecta.

NEAR SIDE *northern hemisphere*

Mare Crisium

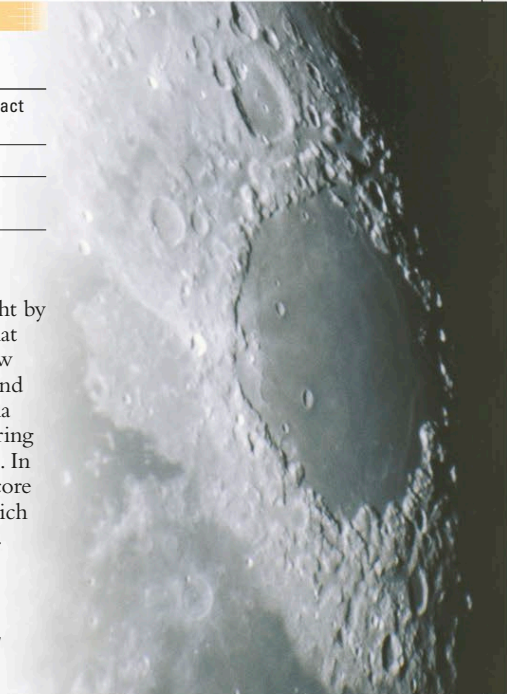


TYPE Lava-filled impact crater (sea)
AGE 3.9 billion years
DIAMETER 350 miles (563 km)

Mare Crisium has an extremely smooth floor, which varies in height by less than 290 ft (90 m). The lava that flooded Crisium had extremely low viscosity and became like a still pond before it solidified. The Soviet Luna 24 probe was the last mission to bring back rock samples from the Moon. In 1976, it returned to Earth with a core of rock weighing 6 oz (170 g), which was collected from Crisium's floor.

OVAL CRATER

The Mare Crisium, which can be seen with the naked eye from Earth, is nearly circular in shape. Over 95 percent of lunar craters are completely circular.



NEAR SIDE *northern hemisphere*

Montes Apenninus



TYPE Mountain range
AGE 3.9 billion years
LENGTH 249 miles (401 km)

The Lunar Apennine mountains form a ring around the southeastern edge of the Mare Imbrium impact basin. They consist of crustal blocks rising more than 1.9 miles (3 km) above the flat lava plain, pushed up by the shock wave from the Imbrium impact. The mountain chain stretches for some 375 miles (600 km), though its southern end is partially buried beneath lava flows.

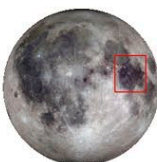
LUNAR MOUNTAINS

The Apennines lie in the lower right of this Apollo 15 image. The dark area to their left is Palus Putredinis.



NEAR SIDE *northern hemisphere*

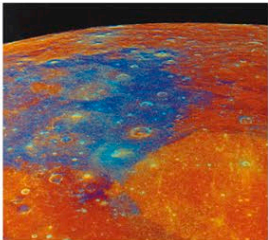
Mare Tranquillitatis



TYPE Sea
AGE 3.6 billion years
DIAMETER 542 miles (873 km)

The surfaces of lunar maria are much darker than highland rock and are also considerably younger. This means that they are relatively smooth and contain only a few impact craters. Their low reflectivity is due to the chemistry of

the very fluid lava that flooded them. The Mare Tranquillitatis (Latin for “Sea of Tranquillity”) lies just north of the lunar equator and joins onto the southeast part of the Mare Serenitatis (Sea of Serenity). Together, the two seas form one of the Moon's most prominent features. The basin in which the “sea” formed is very ancient, predating the formation of the Imbrium Basin 3.9 billion years ago. It overlaps with other basins at



RICH IN TITANIUM

This Galileo image has been color-coded according to the titanium content of the rock. The blue Tranquillitatis region is rich in titanium, whereas the orange Serenitatis region at the lower right is titanium-poor.

several points but only flooded with lava about 3.6 billion years ago. The Sea of Tranquillity was famously the landing place of US astronauts Neil Armstrong and Buzz Aldrin on their 1969 Apollo 11 mission.



BEFORE TOUCHDOWN

The flat, desolate plain of the Sea of Tranquillity stretches away to the north in this view from the Apollo 11 lunar module taken just before landing.